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Degree: B-Tech. Semester 3rd

END-SEMESTER EXAMINATION, June, 2026

Course Title: Probability Theory and Random Process
Course Code: Sem: ECECC06

Duration: 3.00 Hours
Max. Marks: 40

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

<table>
<tr><th>Q. No.</th><th>Question</th><th>Marks</th><th>CO</th></tr>
<tr><td>1.</td><td>Attempt any two parts of the following:</td><td></td><td></td></tr>
<tr><td>1a</td><td>Let X be a continuous random variable with the following distribution:
$$f(x) = \begin{cases} kx, & if\ 0 \le x \le 5 \\ 0, & elsewhere \end{cases}$$
(a) Evaluate k. (b) Find $P(1 \le X \le 3)$</td><td>4</td><td>C01</td></tr>
<tr><td>1b</td><td>Let X and Y be independent random variables. Then show that
(i) E[XY]=E[X] E[Y] ,
(ii) Var(X+Y) = Var(X)+Var(Y),</td><td>4</td><td>C01</td></tr>
<tr><td>1c</td><td>Find the expectation (μ), variance (σ^2), and standard deviation (σ) of the following distribution
<table><tr><td>x_i</td><td>2</td><td>3</td><td>11</td></tr><tr><td>$f(x_i)$</td><td>$\frac{1}{3}$</td><td>$\frac{1}{2}$</td><td>$\frac{1}{6}$</td></tr></table></td><td>4</td><td>C01</td></tr>
<tr><td>2.</td><td>Attempt any two parts of the following:</td><td></td><td></td></tr>
<tr><td>2a</td><td>Let X and Y be random variables with join distribution given below:
<table><tr><td>X \ Y</td><td>−3</td><td>2</td><td>4</td><td>Sum</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.5</td></tr><tr><td>3</td><td>0.3</td><td>0.1</td><td>0.1</td><td>0.5</td></tr><tr><td>Sum</td><td>0.4</td><td>0.3</td><td>0.3</td><td></td></tr></table>
Find: (a) the distributions of X and Y, (b) $Cov(X,Y)$, (c) $\rho(X,Y)$</td><td>4</td><td>C02</td></tr>
<tr><td>2b</td><td>Team A has probability 2/3 of winning whenever it plays. If A plays 4 games, find the probability that A wins (i) exactly 2 games, (ii) at least 1 game, (iii) more than half of the games.</td><td>4</td><td>C02</td></tr>
<tr><td>2c</td><td>Let X be a random variable with the Poisson distribution $p(k,\lambda)$, Then show that (i) E(X)= λ and (ii) Var (X)= λ.</td><td>4</td><td>C02</td></tr>
<tr><td>3.</td><td>Attempt any two parts of the following:</td><td></td><td></td></tr>
<tr><td>3a</td><td>State and prove Chebyshev inequality</td><td>4</td><td>C03</td></tr>
<tr><td>3b</td><td>Let X be a random variable with the following distribution and let $Y = X^2$, Determine (i) the distribution (g) of Y (ii) the joint distribution (h) of X and Y.</td><td>4</td><td>C03</td></tr>
</table>

x_i	-2	-1	1	2
$f(x_i)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

3c	The joint PDF of the random variables X and Y is given as $f_{XY}(x,y) = \frac{1}{4}e^{-\lvert x\rvert-\lvert y\rvert}, -\infty < x < \infty, -\infty < y < \infty$ Determine (i) Whether the random variables X and Y are statistically independent, (ii) The probability that $X \leq 1$ and $Y \leq 0$.	4	C03
4.	*Attempt any two parts of the following:*		
4a	Define the following terms in detail : (i) Stationary random Process (ii) Ergodic process	4	C04
4b	A random process is expressed as $X(t) = A\cos(\omega t + \theta)$, where, ω and θ are constants and A is a random variable. Determine whether X(t) is wide sensed stationary WSS process.	4	C04
4c	If the random process is wide sense stationary than prove that $R_{XX}(-\tau) = R_{XX}(\tau)$	4	C04
5.	*Attempt any two parts of the following:*		
5a	Derive the expression for power spectral density of a wide-sensed process.	4	C05
5b	Define the autocorrelation function of a random process, derive its expression and explain its significance.	4	C05
5c	Discuss cross correlation and Gaussian process in detail.	4	C05